

Land Survey

# Engineering Mathematics I

## Revision Questions

Use these questions for CAT and summative assessment preparation.

## Formula Practice

1. Apply the formula below in a technical example.

$$d = \sqrt{(\Delta E)^2 + (\Delta N)^2}$$

2. Apply the formula below in a technical example.

$$\theta = \tan^{-1} \left( \frac{\Delta E}{\Delta N} \right)$$

3. Apply the formula below in a technical example.

$$\sin \theta = \frac{O}{H}$$

4. Apply the formula below in a technical example.

$$\cos \theta = \frac{A}{H}$$

## Applied Practice

5. Find distance between two survey points.
6. Calculate bearing from coordinate differences.
7. Use trigonometry to calculate height.